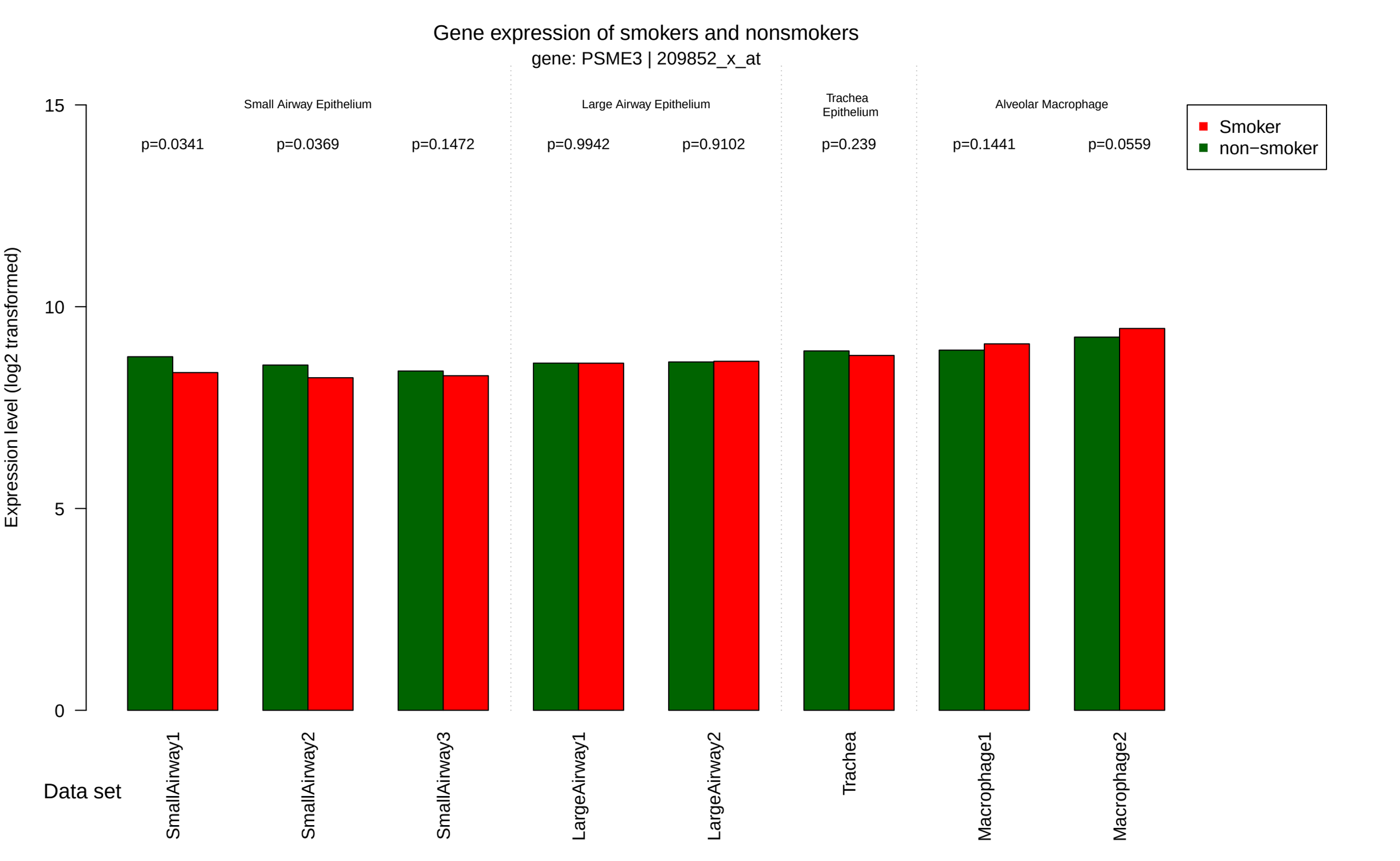
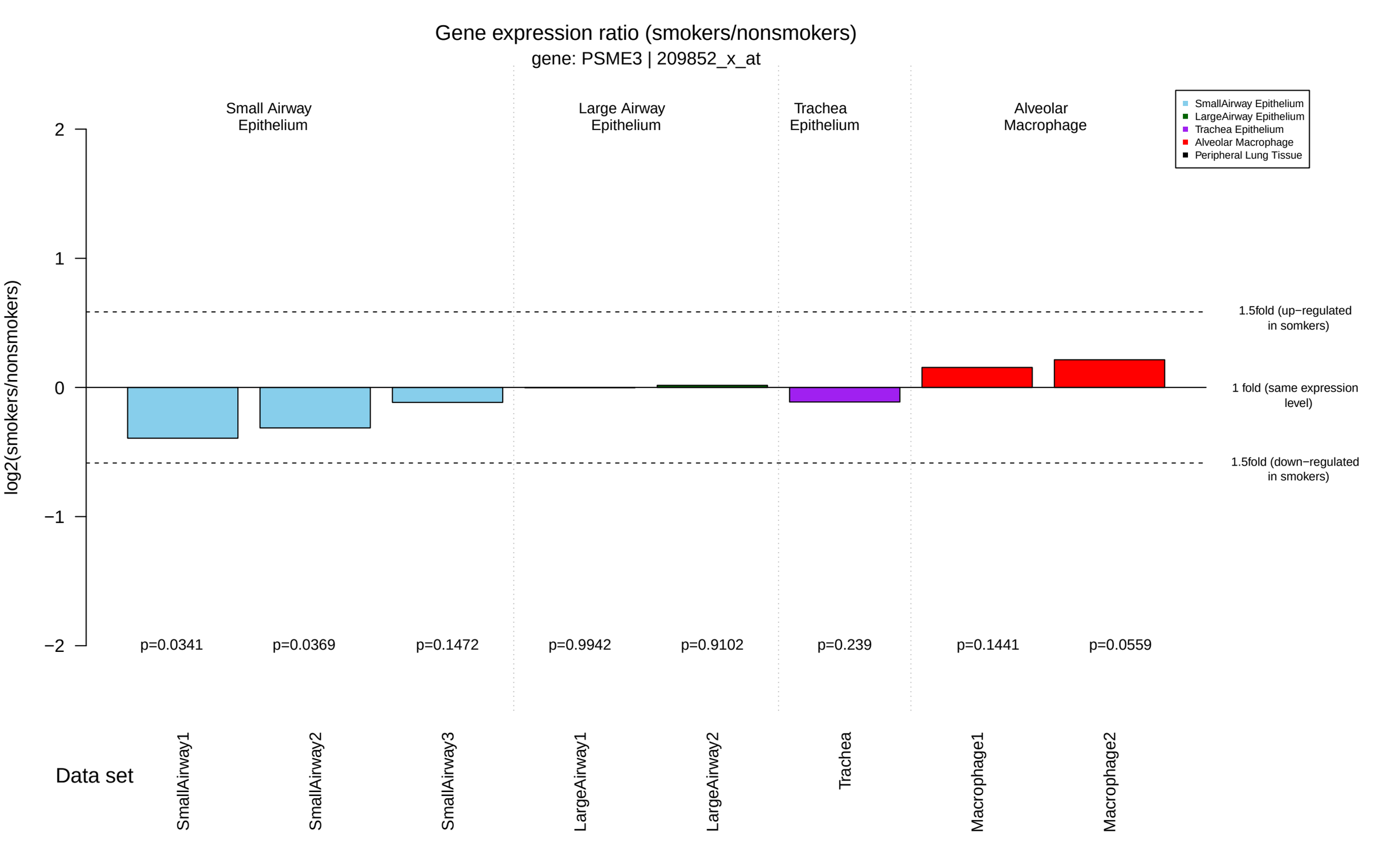
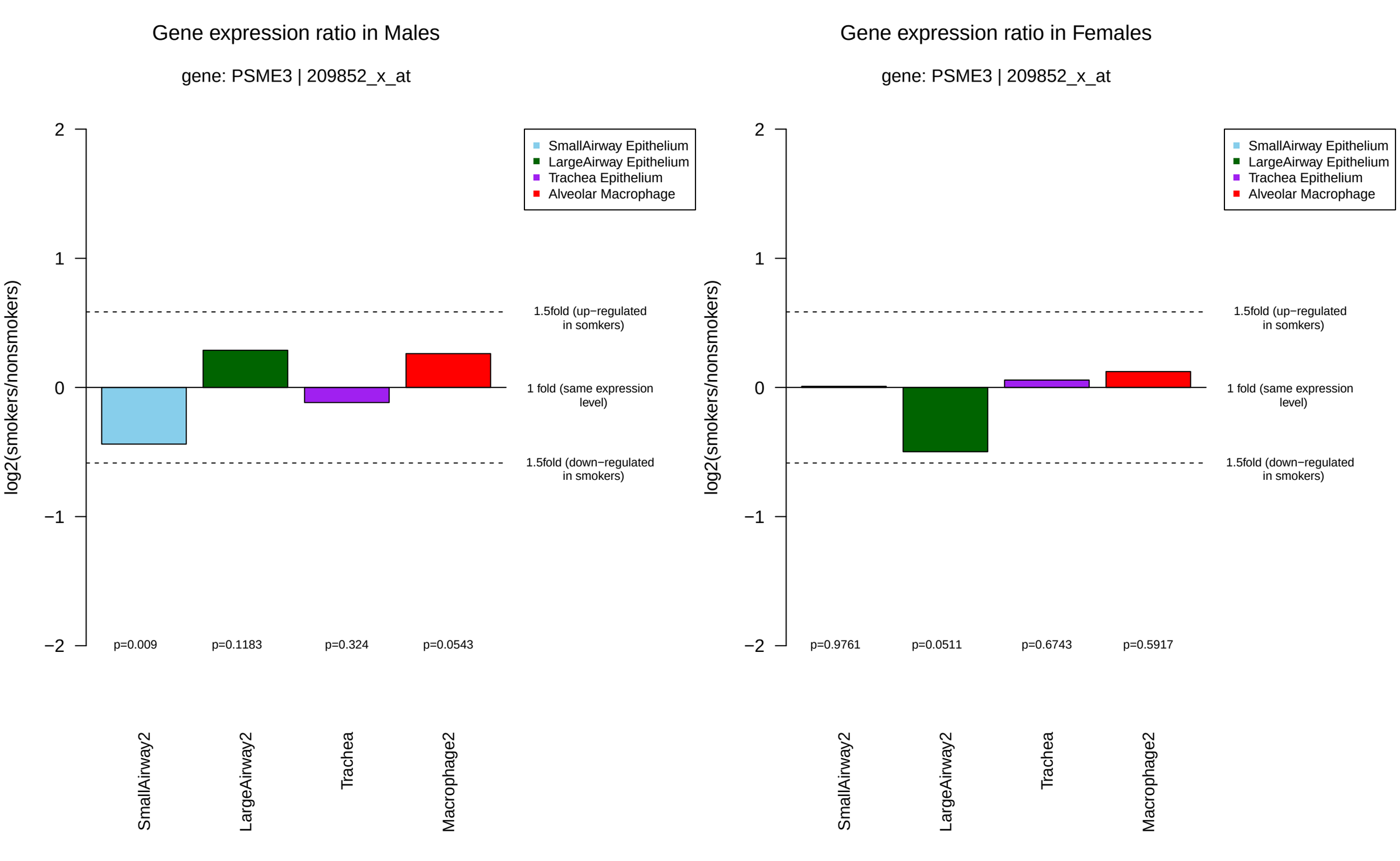
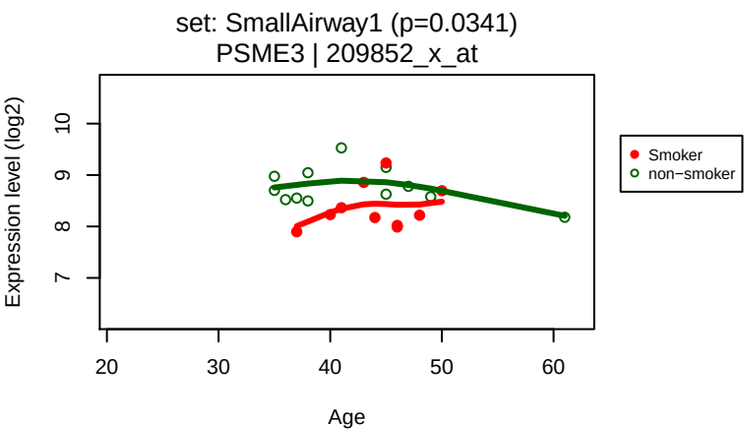
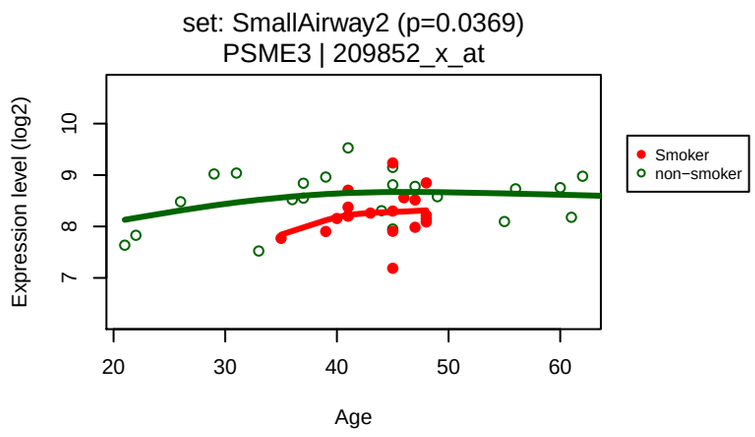
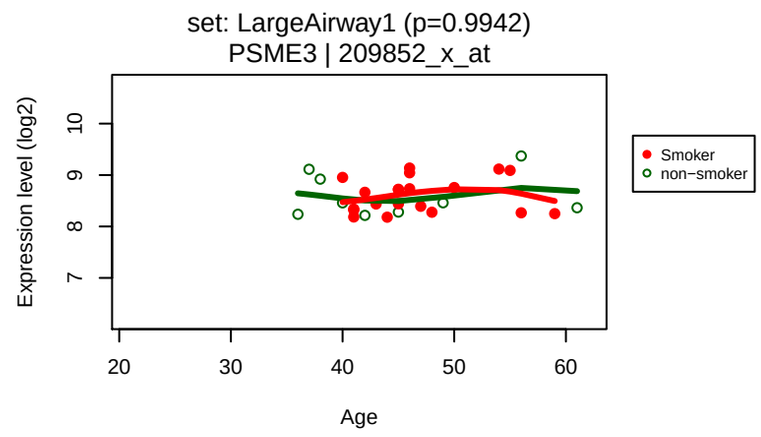

Gene expression of smokers and nonsmokers
gene: PSME3 | 209852_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0341
p=0.0369
p=0.1472
p=0.9942
p=0.9102
p=0.239
p=0.1441
p=0.0559
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

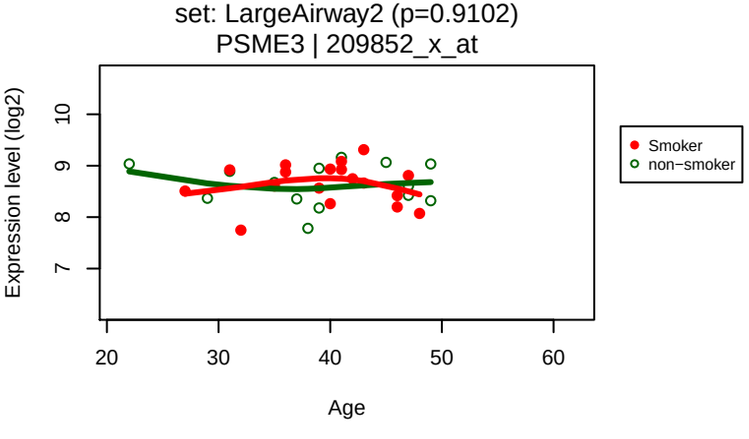

Gene expression ratio (smokers/nonsmokers)
gene: PSME3 | 209852_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0341
p=0.0369
p=0.1472
p=0.9942
p=0.9102
p=0.239
p=0.1441
p=0.0559
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

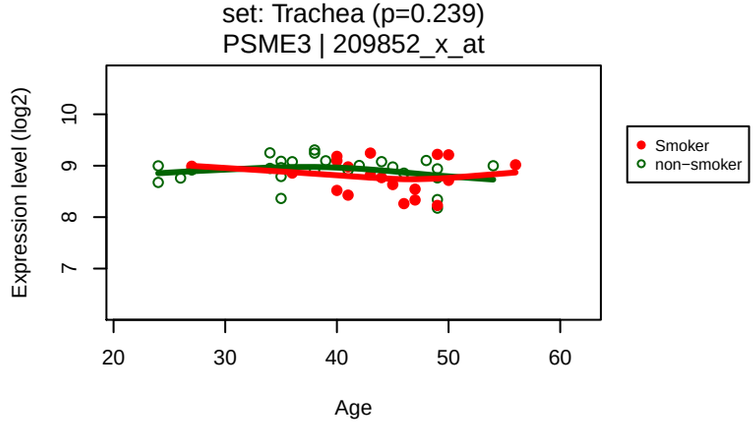

Gene expression ratio in Males
gene: PSME3 | 209852_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.009
p=0.1183
p=0.324
p=0.0543
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSME3 | 209852_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9761
p=0.0511
p=0.6743
p=0.5917
SmallAirway2
LargeAirway2
Trachea
Macrophage2

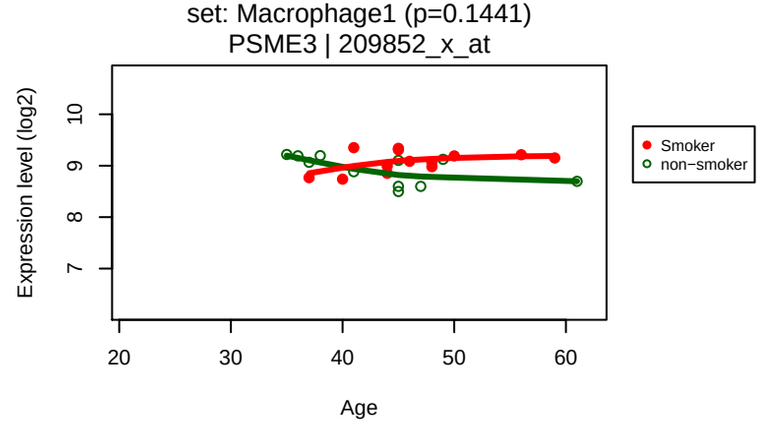

set: SmallAirway1 (p=0.0341)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker

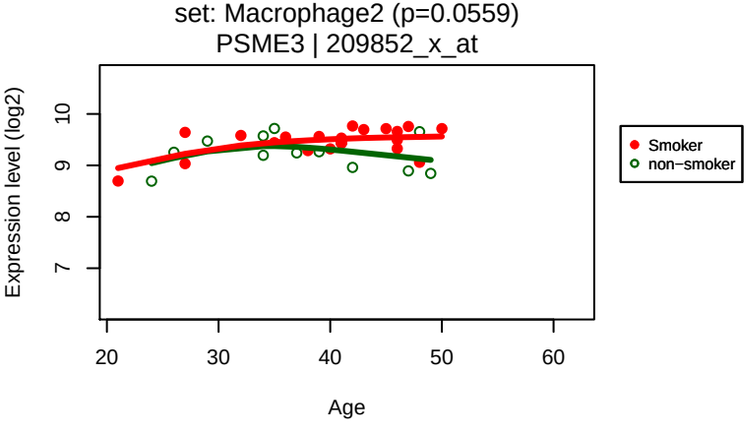

set: SmallAirway2 (p=0.0369)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker

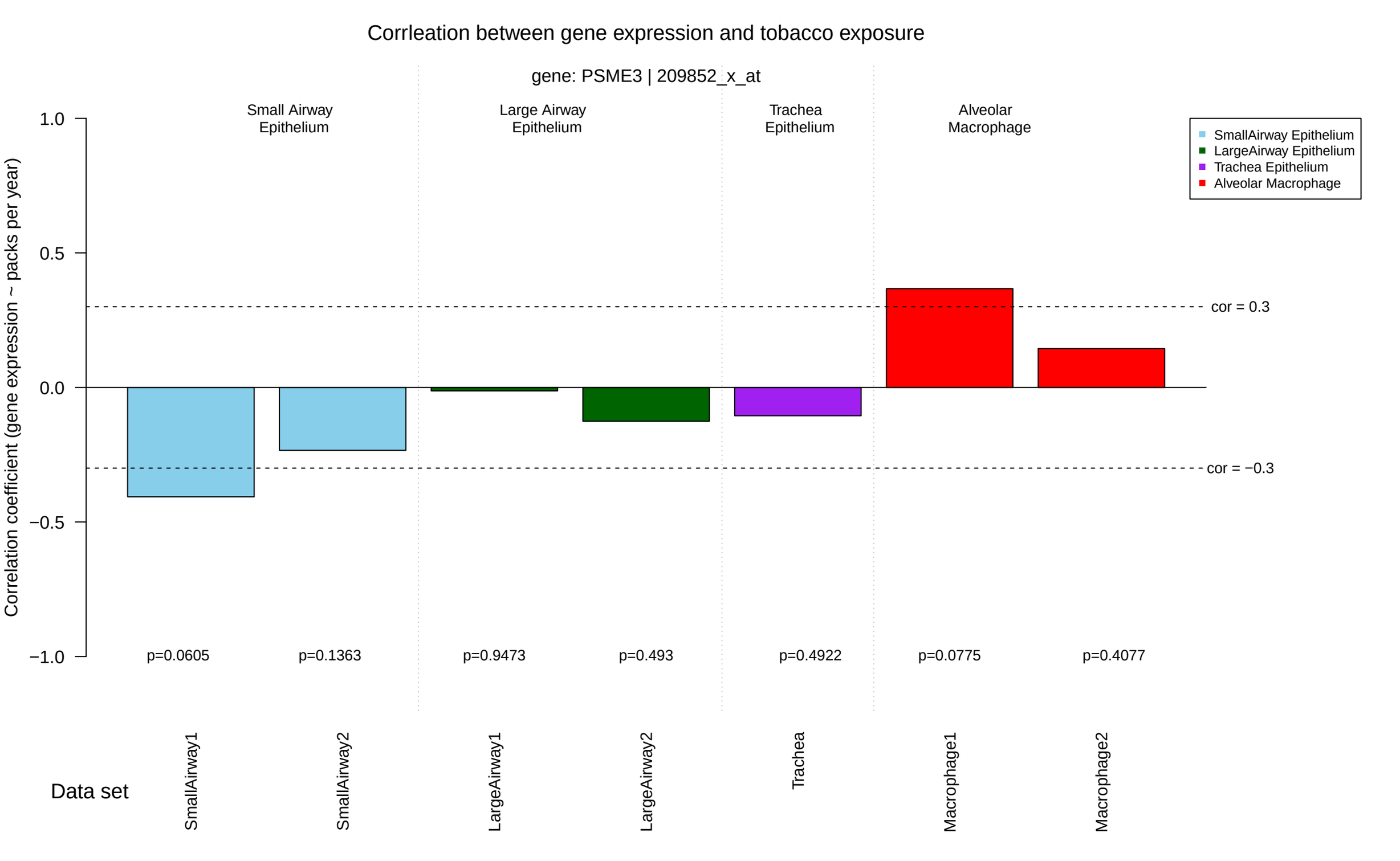

set: LargeAirway1 (p=0.9942)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9102)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.239)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1441)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0559)
PSME3 | 209852_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSME3 | 209852_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0605
p=0.1363
p=0.9473
p=0.493
p=0.4922
p=0.0775
p=0.4077
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2